PID Controller Turning and Response Analysis

```
clc;
clear;
close all;

% Plant Transfer Function
% G(s) = 1/(s^2 + 10s + 20)

% PID Controller Parameters
Kp = 30;
Ki = 70;
Kd = 2;

% Simulation Time
tspan = 0:0.01:10;

% Initial Conditions
% x(1) = Output
% x(2) = Velocity
% x(3) = Integral of Error
x0 = [0; 0; 0];

% PID Controlled System Equations
rhs = @(t,x) [
```

```
    x(2);
    Kp*(1-x(1)) + Ki*x(3) ...
    - (10+Kd)*x(2) - (20+Kp)*x(1);
    1-x(1)
];

% Solve Differential Equations
[t,x] = ode45(rhs,tspan,x0);

% System Output
y = x(:,1);

% Plot Step Response
figure(1);
plot(t,y,'b','LineWidth',2);
hold on;
plot(t,ones(size(t)),'r--','LineWidth',1.5);
grid on;
xlabel('Time (seconds)');
ylabel('Amplitude');
title('PID Controlled System Response');
legend('PID Response','Reference Input');

% Performance Analysis
steady_value = y(end);
```

```
[peak_value,peak_index] = max(y);
peak_time = t(peak_index);

overshoot = max(0,(peak_value-1)*100);

% Rise Time Calculation
i10 = find(y>=0.1,1);
i90 = find(y>=0.9,1);

if ~isempty(i10) && ~isempty(i90)
   rise_time = t(i90)-t(i10);
else
   rise_time = NaN;
end

% Settling Time Calculation (2%)
idx = find(abs(y-1)>0.02,1,'last');

if isempty(idx)
   settling_time = 0;
elseif idx < length(t)
   settling_time = t(idx+1);
else
   settling_time = NaN;
end
```

```
% Display Performance Results
disp('PID PERFORMANCE PARAMETERS');
disp('---------------------------');

fprintf('Steady State Value = %.4f\n',steady_value);
fprintf('Peak Value = %.4f\n',peak_value);
fprintf('Peak Time = %.4f seconds\n',peak_time);
fprintf('Overshoot = %.2f %%\n',overshoot);
fprintf('Rise Time = %.4f seconds\n',rise_time);
fprintf('Settling Time = %.4f seconds\n',settling_time);

hold off;
```

Simulink

